Manual

Licence plate detection with Jetson Nano

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Acronyms

1 Getting Started

Welcome to the Smart License Plate Detection System

Dear User,

Welcome to your **Smart License Plate Detection System** — a reliable ML-powered solution developed at **Hochschule Emden–Leer**. This device uses advanced edge computing on the **NVIDIA Jetson Nano** to automatically recognise and log vehicle license plates in real time.

The system starts automatically when powered on — no setup or configuration is required. Within seconds, it begins detecting and displaying vehicles on screen with accurate entry, exit, and duration details.

Key Highlights:

- Works entirely offline — no internet required.
- Designed for 24/7 airport gate operation.
- Automatically handles billing for stays beyond 15 minutes.
- Recognises exempt vehicles (police, ambulance, government).
- Compact, efficient, and easy to maintain.

"Technology that simplifies — accuracy that empowers."

Developed by **Gautam Ramesh**, **Siddhanth Sharma**, and **Shubhankar Dumka**

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Before You Start

- Place the device on a stable, flat surface with good ventilation.
- Make sure the **power supply (5 V / 4 A)** is connected firmly and the cable is not loose.
- Keep the **camera lens clean** for accurate plate reading.

- Do not block the openings on the aluminum case; it keeps the system cool.
- Avoid exposing the device to rain, moisture, or direct sunlight.
- Connect the HDMI cable to your monitor before switching on the power.
- Wait until the indicator light turns on — this means the system is ready.
- If the display stays blank, check power and camera connections first.

These simple steps help ensure your system starts safely and performs reliably every time.

Important to Know Before Use

- This system records number plates for time tracking and billing at the airport gate.
- Do not move or touch the camera once it is aligned. Misalignment can cause unreadable plates.
- If a plate is too dirty or not readable, you may need to type it in manually later.
- Emergency and official vehicles (police, ambulance, etc.) are recognised as **no charge**.
- Turn off the system only after normal shutdown. Unplugging power while it is running can cause data loss.

For detailed instructions on setup and use, please refer to chapter 5.

These simple steps help ensure your system starts safely and performs reliably every time.

2 Main Function

2.1 Main Function

The **Airport License Plate Detection System** is a smart camera-based solution that automatically recognises and records the license plates of vehicles entering and leaving the airport area. It is specially designed for **German license plates** and helps manage short-term parking at the airport gate efficiently and fairly. The device uses the compact and reliable **NVIDIA Jetson Nano** computer, which performs all detection, timing, and billing functions locally — without requiring any internet or cloud connection [NVI24].

When a vehicle enters, the camera captures the license plate and stores it together with the date and time. When the same vehicle exits, the system detects it again and automatically calculates how long it stayed inside the airport area. If the duration is more than **15 minutes**, the system marks it for billing. If the stay is **15 minutes or less**, the parking is free of charge [PM23].

Certain vehicles such as **emergency services (ambulance, police, fire brigade)** and **official government vehicles** are automatically identified and exempted from all charges. These special plates are pre-registered in the system's database, ensuring that authorised vehicles always pass freely without manual intervention.

All vehicle data, including plate number, entry time, exit time, duration, and payment status, are securely stored inside the device. This ensures transparency, traceability, and smooth traffic flow at both the entry and exit points of the airport [Ult23; Jai21].

Key Highlights:

- Recognises and records German license plates automatically.
- Calculates parking duration and applies the 15-minute free rule.
- Automatically exempts emergency and government vehicles from charges.
- Works fully offline — no internet or network setup required.
- Stores all records locally for reporting, auditing, or payment.

2.2 How the System Works

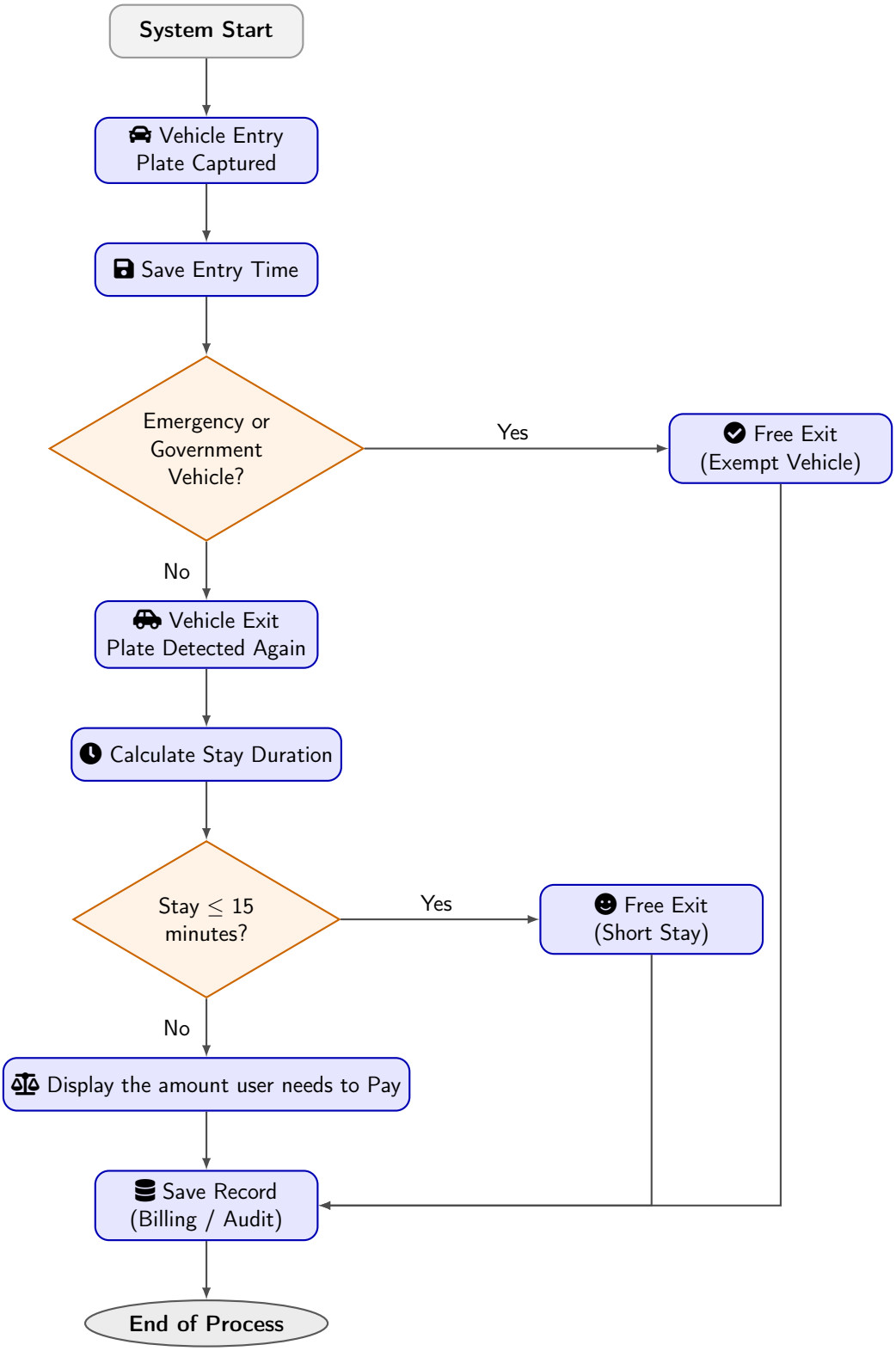

Figure 2.1: Final airport entry–exit workflow with exemptions, duration check, and billing logic.

The system repeats this workflow continuously throughout the day. By automatically checking for exempt vehicles and applying the 15-minute free rule, it guarantees fairness, speed, and efficiency for all users.

[Future Image Placeholder]

Concept image of the airport setup will be added here.

Figure 2.2: Placeholder: Concept image showing system deployment.

The system is built to work smoothly and automatically with no special setup. Once the device is connected to power, it starts on its own and is ready for use within a few seconds. The camera watches both the entry and exit lanes, while the screen shows every vehicle that passes — along with how long it stayed and whether any charge applies.

All processing happens inside the compact Jetson Nano computer. There is no need for an internet connection or external software. The system is designed to be reliable even during continuous, all-day operation.

Camera Placement and Setup:

- Mount the camera about **2.0–2.5 metres above ground level**, facing the vehicle lane at a **30–40° angle**.
- Keep a distance of about **3–5 metres** between the camera and the point where vehicles stop or slow down.
- Avoid placing the camera directly opposite bright lights or reflective surfaces.
- Ensure there is enough ambient light. During night hours, add a small **LED lamp or floodlight** near the gate for better clarity.
- Keep the camera lens clean and free of dust, water spots, or scratches.
- Do not position the camera behind glass or near strong sunlight to prevent reflections.

Plausibility and Manual Verification:

- If the system cannot read a plate clearly (for example, due to dirt, glare, or motion blur), it will display a **"Detection Unclear"** message on the screen.
- In such cases, the operator can manually enter the license plate number using a simple on-screen input field or log sheet.
- The system checks for **duplicate or incomplete entries** to avoid double billing.
- If an entry or exit is missing, the software automatically flags the record for review.
- For fairness, vehicles with unreadable plates are temporarily marked as **"Pending Verification"** until corrected by staff.
- Regular cleaning of the lens and maintaining good lighting reduces the chance of detection errors.

In simple terms:

- Plug in the power — the system starts automatically.
- The camera reads license plates at entry and exit points.
- The display shows vehicle timing and free/charge status.
- Emergency or government vehicles are recognised and never charged.
- In rare cases of unclear detection, staff can correct the record manually.
- Everything runs locally — no internet or cloud connection required.

—

2.3 User Interaction

The system is designed for daily airport use with minimal user effort. All detections, timings, and charge decisions appear automatically on the display — no keyboard or technical input is required. The operator only monitors the live feed and confirms that vehicles are captured correctly.

Interface Overview:

- The screen displays the **camera view** on the left and the **detection log** on the right.
- Each entry shows the plate number, time, duration, and parking status — "Free" (green) or "Charge" (orange).

- Exempt vehicles such as police or ambulance appear in blue with "No Charge."

User Feedback and Manual Input:

- A short sound confirms each successful detection.
- If a plate is unclear, the alert *"Detection Unclear – Please Verify"* appears.
- The operator can type the correct number manually; it is saved instantly as a verified record.

Managing Exempt Vehicles:

- Exempt vehicle numbers are stored in a simple text file (`exempt_list.txt`).
- Staff can edit this list before startup; changes take effect immediately.

Daily Operation:

- Power on — the system starts automatically.
- Observe the live feed and alerts.
- Update exempt list when needed.
- Turn off display and power after working hours; all data is saved automatically.

This workflow ensures clear, automatic, and reliable operation for non-technical staff, focusing on smooth vehicle flow and accurate billing.

3 System Sketch

3.1 System Overview

The following sketch gives a simple view of how all parts of the **License Plate Detection System** are connected. Each device works together automatically from the moment the power is switched on — to detect, read, and record vehicle license plates at the airport gate.

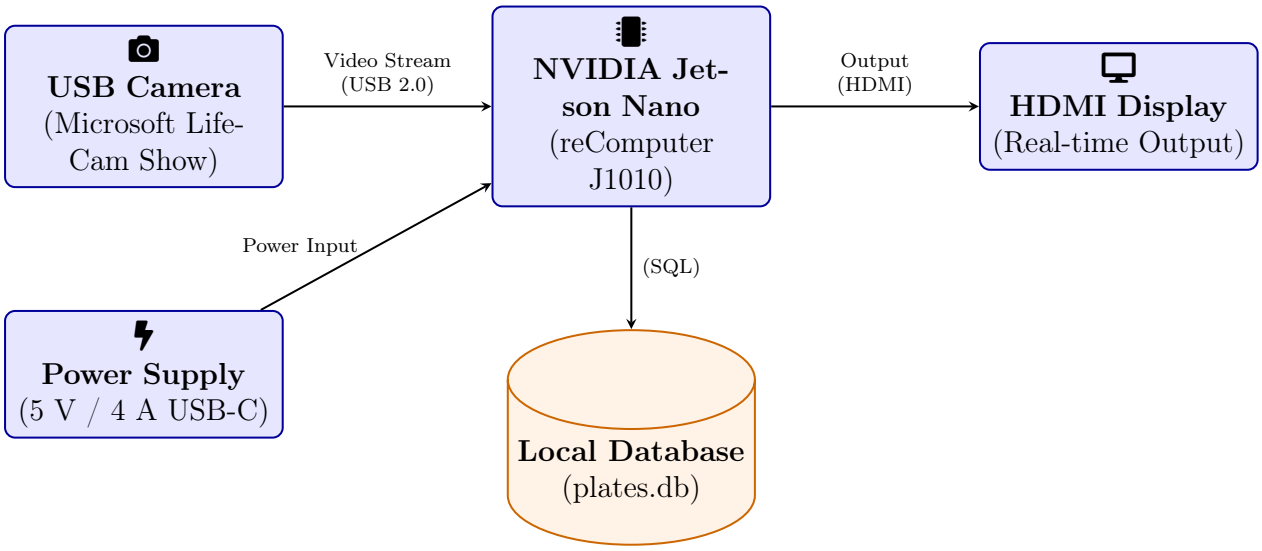

Figure 3.1: Functional hardware sketch of the Jetson Nano License Plate Detection System

3.2 Hardware Components

Each part of the system plays a specific role in making automatic plate recognition work smoothly:

- **Camera:** Captures live video of vehicles entering or exiting the airport gate.
- **Jetson Nano (reComputer J1010):** The main unit that reads the video, detects number plates, and calculates parking time.
- **Display Screen:** Shows live detection results and timing information.

- **Power Adapter:** Supplies stable 5V/4A power through the USB-C connector.
- **Local Database:** Stores all plate numbers with date, time, and charge status.

3.3 System Summary

Component	Purpose	Description
Camera	Captures video	Microsoft LifeCam Show USB camera fixed near gate.
Jetson Nano	Processes video	Small AI computer that detects and reads plates.
Display	Shows results	HDMI monitor displaying detected plates in real time.
Database	Stores records	Saves each plate with time, date, and status (free or charged).
Power Supply	Runs the system	5V / 4A USB-C adapter powering the Jetson Nano and camera.

Table 3.1: System components and their basic roles.

The system is compact, quiet, and easy to set up. Once powered, it runs automatically and provides a clear, continuous view of all vehicle detections — ensuring accurate, fair, and paperless parking management.

4 List of Parts

4.1 Overview

This section lists all major components used in the **Airport License Plate Detection System**. Every part is selected for reliability, ease of replacement, and full compatibility with the **NVIDIA Jetson Nano** platform.

Table 4.1: Main Components of the System

Part Name	**Description / Function**	**Model / Type**	**Qty.**
Main Unit	Jetson Nano 4GB (reComputer J1010) – runs detection, timing, and billing	Seeed Studio / NVIDIA	1
Camera	Captures live vehicle images at the airport gate	Microsoft LifeCam Show (USB 2.0)	1
Camera Stand	Adjustable metallic stand for stable positioning	Generic Steel Mount	1
Display	HDMI monitor showing real-time detections	15.6-inch / 1080p	1
Storage	64 GB microSD card with JetPack OS	SanDisk / Samsung Evo+	1
Power Supply	5V / 4A USB-C adapter for Jetson Nano and camera	Seeed Studio Certified	1
Ethernet Cable (Optional)	Used only for setup, updates, or remote access – not required for daily use	Cat 6 / 1.5 m	1
Software Tools	YOLOv8 (detection), EasyOCR (reading), SQLite (logging)	Python-based open source stack	–

Each component is plug-and-play, making the system simple to assemble, transport, and maintain. The setup is compact, energy-efficient (under 10 W), and fully operational without internet access — ideal for continuous airport use.

5 Set Up

5.1 System Setup

This section explains, in simple steps, how to prepare and start the **License Plate Detection System**. No technical knowledge is required — follow the setup steps below to get the system ready for use.

Step 1: Place and Connect the Hardware

- Place the **Jetson Nano (reComputer J1010)** on a stable, well-ventilated surface.
- Mount the **camera** on its stand and point it toward the vehicle lane or entry gate.
- Connect the camera to the Jetson Nano using the USB port.
- Connect the **HDMI cable** from the Jetson Nano to the monitor.
- Plug in the **power adapter (5V / 4A USB-C)** but do not turn it on yet.

Step 2: Start the System

- Insert the **microSD card** (already loaded with JetPack 5.1.2).
- Turn on the power switch.
- Wait a few seconds for the system to start automatically.
- The display will show the live camera view once it is ready.

Step 3: Verify Operation

- Watch the screen — detected number plates will appear automatically.
- Adjust the camera height or tilt if the plate is not clearly visible.
- Ensure good lighting at the vehicle gate area. For accurate plate reading, maintain an illumination level between **250 lux (minimum)** and **900 lux (maximum)** — with an optimal range of **400–700 lux**. Avoid shining lights directly into the camera to prevent glare or overexposure.

- The system saves every detection with date and time automatically.

Step 4: During and After Use

- The system runs continuously and does not need supervision.
- After use, simply turn off the power supply.
- All data is stored safely for later review or reporting.

System Setup Diagram:

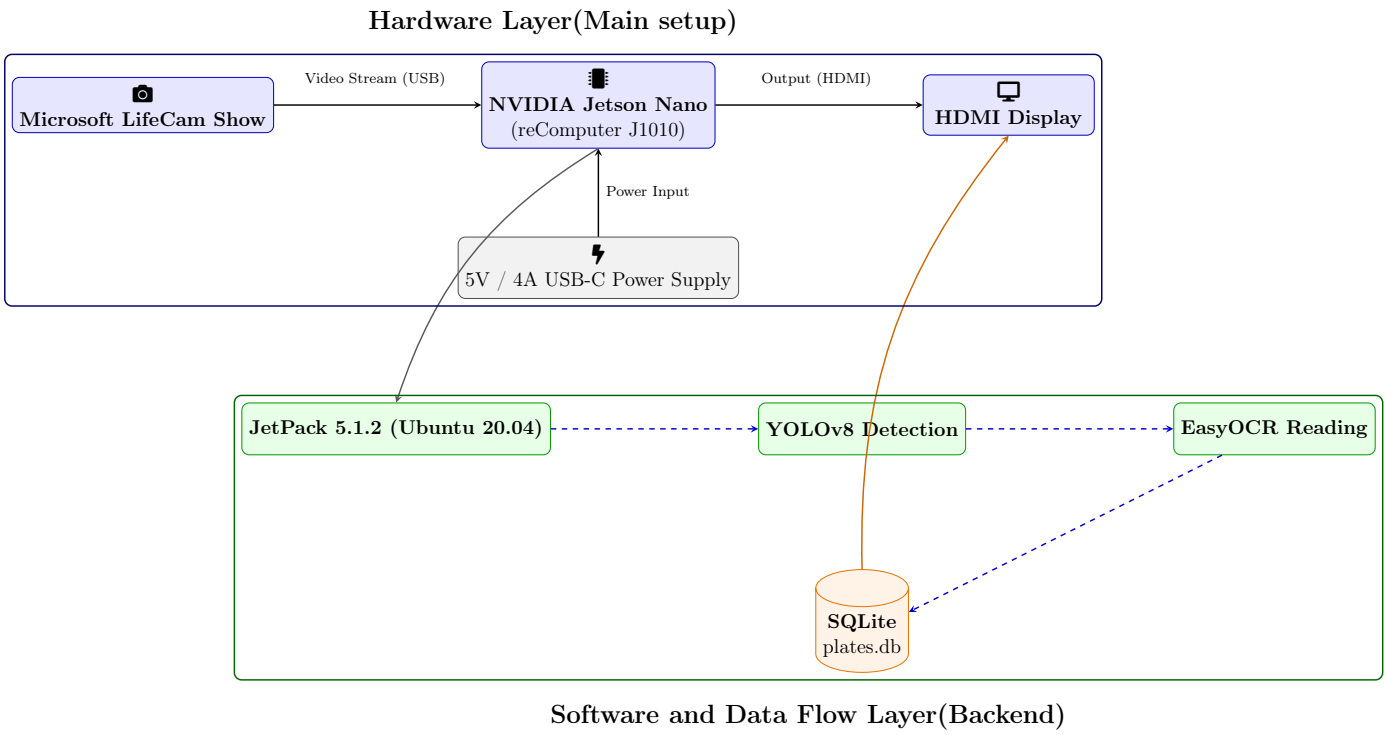

Figure 5.1: System setup overview – hardware and software working together.

Note: The upper part of Figure 5.1 shows the **outer hardware setup**

The lower part represents the **internal processing layer**, which runs automatically inside the Jetson Nano. It includes the system software, detection model, and database used for number plate recognition and data storage. This section is mainly relevant for developers who want to understand or extend the system's logic.

6 Maintenance

6.1 Maintenance

Regular maintenance keeps the **License Plate Detection System** running smoothly and extends its lifetime. This section describes how users can care for the hardware and software to ensure reliable daily operation at the airport gate.

6.1.1 Hardware Maintenance

- **Keep it clean:** Gently wipe the Jetson Nano unit and camera with a dry microfiber cloth. Avoid liquids, direct sunlight, or dust build-up around the device.
- **Check connections:** Ensure all USB, HDMI, and power cables are firmly attached. Loose connections can cause flickering displays or camera interruptions.
- **Maintain good cooling:** Place the system in a ventilated area. If the device feels unusually hot, turn it off for a few minutes or use a small desk fan.
- **Power stability:** Always use the original 5V/4A USB-C power adapter. Unstable power may lead to sudden restarts or data loss.
- **Camera alignment:** Check every few weeks that the camera faces the correct lane and captures clear plates without obstruction.

6.1.2 Software Maintenance

- **Regular check-ups:** Restart the system once every few days to clear memory and refresh processes.
- **Keep the system updated:** If an update notification appears, allow the Jetson Nano to install recommended system updates.
- **Database management:** Every few weeks, check that the detection log is not full. If storage is low, export or delete older records safely.
- **Model refresh (for developers):** Advanced users can retrain or update the detection model to improve recognition accuracy over time.

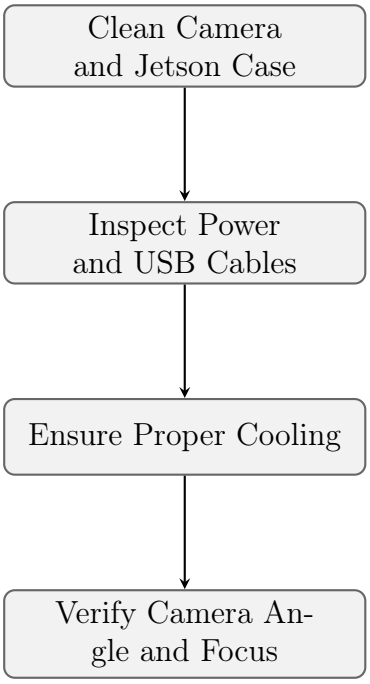

Figure 6.1: Routine hardware maintenance flow.

6.1.3 Backup and Safety

- **Weekly backup:** Copy the database file (`plates.db`) to a USB drive or computer for safekeeping.

- **Power off correctly:** Always shut down the device through the desktop before unplugging power.

- **Avoid sudden shutdowns:** Interrupting the system during detection can damage stored data.

- **Emergency reset:** If the screen freezes, disconnect power, wait 10 seconds, and reconnect.

- **Check storage space:** Ensure there is enough free space on the Jetson Nano's microSD card. Low storage can slow down performance or prevent new data from being saved.

TODO: Future addition – Backup & Maintenance Dashboard UI mockup.

Figure 6.2: (To be added) Example of weekly backup process.

Table 6.1: Recommended Maintenance Schedule

Task	**Frequency**	**Performed By**
Clean device and camera lens	Monthly	Operator
Check cable connections	Monthly	Operator
Back up database	Weekly	Operator
Update system software	Quarterly	Technician
Inspect camera focus and lighting	Monthly	Operator

6.1.4 Maintenance Schedule

Safety Reminder

- Turn off power before cleaning or adjusting cables.
- Avoid using sharp objects or sprays near the camera lens.
- Do not block the ventilation openings of the Jetson Nano case.
- If the device overheats or smells unusual, disconnect power immediately and contact support.

Summary: Routine maintenance keeps the system stable, accurate, and long-lasting. With regular cleaning, safe power use, and simple backups, the License Plate Detection System will continue operating reliably every day.

7 Safety Guidelines

7.1 Safety Guidelines

The following safety guidelines are essential to ensure the reliable, secure, and fault-free operation of the **License Plate Detection System**. They cover electrical, thermal, mechanical, and operational aspects relevant to the embedded AI hardware setup based on the NVIDIA Jetson Nano.

7.1.1 Electrical Safety

- Always use a certified **5V / 4A USB-C power supply** with built-in safety protection against overcurrent and short-circuits.
- Avoid connecting or disconnecting peripherals (camera, HDMI, USB) while the system is powered on.
- Check all cables for damage or wear before use, and make sure they are properly insulated.
- Use grounded power outlets or a surge protector to prevent voltage spikes or static discharge.
- Connecting cables incorrectly may cause permanent damage to the board or its peripherals.
- Avoid operating the system outdoors during rain, lightning, or unstable weather conditions.

7.1.2 Thermal and Mechanical Safety

- Ensure good ventilation around the aluminum case — do not block any openings, as the system relies on passive cooling.
- Keep the device within a safe temperature range (**0 °C to 40 °C**). In hotter environments, use a small fan to maintain airflow.
- Keep liquids, food, or heavy objects away from the unit. Avoid high humidity environments.

- Mount the camera securely and ensure it cannot fall or shift during operation.
- When handling the Jetson Nano board directly, use anti-static precautions (e.g., an ESD wrist strap).

7.1.3 Operational Safety

- Always shut down the system properly using software commands (e.g., `sudo shutdown`) before unplugging power.
- Do not overload the USB ports with high-power accessories or unverified devices.
- Monitor for abnormal heating, flickering, or warning messages — if detected, power off immediately and inspect hardware.
- Avoid sudden reboots or forced shutdowns while detections or database operations are running.
- During cleaning or relocation, ensure the power supply is fully disconnected.

7.1.4 Power and Inference Safety

Continuous AI processing generates both heat and power load. To keep the system safe and stable:

- Periodically check the power draw using a USB power meter or multimeter — it should remain within the adapter's rated limit.
- Avoid keeping the system under heavy load for long hours without monitoring temperature.
- Allow short cooling breaks after extended operation or testing sessions.

Summary

Following these safety rules protects both the hardware and the environment where it operates. Electrical and thermal safety prevent component damage, while operational care ensures stability and uptime. With regular checks, stable power use, and proper handling, the License Plate Detection System can operate safely and efficiently for years in real-world airport conditions.

8 Troubleshooting

8.1 Troubleshooting

This section lists the most common issues that may occur while operating the **License Plate Detection System**, along with their likely causes and solutions. These checks help maintain system stability, reduce downtime, and ensure uninterrupted license plate recognition.

8.1.1 Hardware and Power Issues

- **System does not power on:**
  - Confirm the **5V / 4A USB-C adapter** is properly connected and the LED on the Jetson Nano is lit.
  - Test a different cable or wall socket.
  - Avoid using unpowered USB hubs to supply power.
- **Device overheats or shuts down unexpectedly:**
  - Ensure proper airflow and use a small fan if necessary.
  - Run `sudo tegrastats` to monitor temperature.
  - Keep ambient temperature below **40 °C**.
- **Camera not detected:**
  - Run `ls /dev/video*` to verify detection.
  - Re-insert the camera or test it on another PC.
  - Use a main USB port, not a low-power one.

8.1.2 Software and Model Issues

- **YOLOv8 model fails to load:**
  - Ensure the weight file (`best.pt`) is in the correct directory.
  - Reinstall Ultralytics using `pip install ultralytics -upgrade`.
  - Verify dependencies (`torch`, `onnxruntime`, `cv2`) are installed.

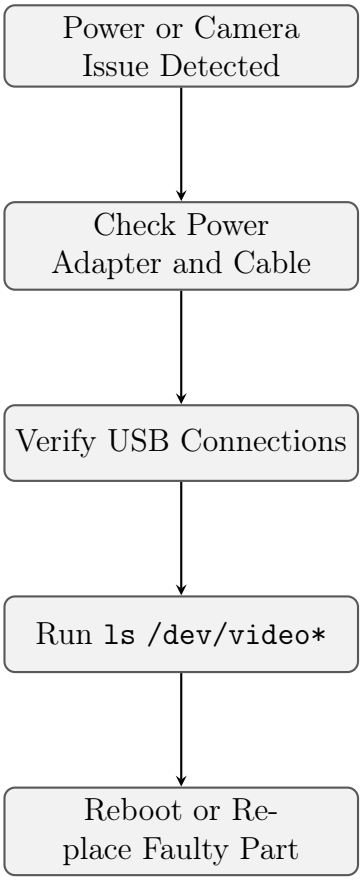

Figure 8.1: Basic hardware troubleshooting flow.

TODO: Add image of Yolov8 load fail.

Figure 8.2: (To be added) Add image of Yolov8 load fail.

- **Low frame rate (FPS):**
  - Use a lighter model (YOLOv8-Nano) or reduce camera resolution to 640×480.
  - Run `sudo nvpmodel -m 0` and `sudo jetson_clocks` for max performance.
  - Close any other GPU-intensive programs.
- **OCR returns wrong text:**
  - Clean the camera lens and ensure good lighting.

  - Adjust image brightness or threshold parameters.
  - Retrain the OCR model with regional plate samples if needed.

8.1.3 Database and Storage Issues

- **Database locked:**
  - Close any parallel process using `plates.db`.
  - Run `sqlite3 plates.db "PRAGMA journal_mode=WAL;"` for safer writes.
  - Restart the app to release file locks.

TODO: Insert image showing database corrupeiont issue.
(Example: screenshot of image showing database corruption issue.)

Figure 8.3: (To be added) Sample troubleshooting interface for real-time error monitoring.

- **Database missing or corrupted:**
  - Restore from backup using `sqlite3 plates.db < backup.sql`.
  - Check free space with `df -h`.
  - Avoid unplugging power during database writes.
- **Storage full:**
  - Delete old logs from `/logs/`.
  - Run `sudo apt clean` to clear unused packages.
  - Expand SD card storage if necessary.

8.1.4 Network or Communication Issues

- **SSH connection fails:**
  - Check IP with `ifconfig` or `hostname -I`.
  - Enable SSH using `sudo systemctl enable ssh`.
  - Inspect router or firewall restrictions.
- **Data not syncing across multiple units:**
  - Confirm all devices have the same network time (`ntpdate`).
  - Test connection via `ping <IP>`.
  - Use a central database or MQTT broker for multi-unit setups.

Troubleshooting Reference Table

Table 8.1: Common Problems, Causes, and Solutions

Problem	Possible Cause	Suggested Solution
No power or LED off	Faulty adapter or cable	Replace adapter and verify connections
Camera not detected	Loose USB or bad port	Reconnect or test on another PC
Model not loading	Missing weight file	Re-download or verify file path
Low FPS	High resolution / throttling	Reduce resolution / enable performance mode
Database locked	Concurrent access	Restart app / enable WAL mode
OCR errors	Poor lighting or angle	Improve illumination / refocus camera
System overheating	Blocked ventilation	Add small fan or reposition device

Summary

Most issues can be resolved by checking power, connections, software dependencies, and database integrity. Using diagnostic commands such as `tegrastats`, `lsusb`, and `sqlite3` simplifies fault identification. Following these troubleshooting steps ensures continuous, reliable, and safe operation of the License Plate Detection System in real-world conditions.

Contact for Technical Assistance

For any technical questions or troubleshooting support, please contact the development team:

- Gautam Ramesh – gautam.ramesh.1@stud-hs-emden-leer.com
- Siddhanth Sharma – siddhanth.sharma@stud.hs-emden-leer.de
- Shubhankar Dumka – shubhankar.dumka@stud.hs-emden-leer.de

Response time: typically within **2–3 working days**.

9 Add Ons

9.1 Proposed Add-Ons

The following add-ons can enhance functionality, usability, and scalability of the **License Plate Detection System**. These modules are optional extensions that allow the system to evolve from a standalone edge device into a smart, networked platform for advanced parking, surveillance, or tolling applications.

- **LoRa or Wi-Fi Module for Remote Access:** Enables wireless communication between multiple Jetson units or a central dashboard. Modules such as `ESP32` or `Seeed Wio-E5 LoRaWAN` can provide long-range connectivity without relying on wired Ethernet.
- **Edge AI Model Optimization:** Integrate **TensorRT** or **NVIDIA DeepStream SDK** to accelerate YOLOv8 inference and reduce latency under heavy load conditions.
- **Cloud Synchronization:** Enable secure backup and analytics by uploading detections and logs to cloud services such as **AWS S3**, **Firebase**, or **Azure IoT Hub**. This supports remote monitoring and historical trend analysis.
- **Automated Billing and Access Control:** Integrate the database with **RFID readers**, **QR code scanners**, or automatic **barrier gates**. This can automate entry/exit billing in parking or restricted zones.
- **Thermal Monitoring Sensor:** Attach a **DS18B20** or **MLX90614** sensor to track the Jetson Nano's internal temperature. This allows automatic activation of a cooling fan if temperature exceeds safe thresholds.
- **Solar-Powered Operation:** Deploy the entire setup on a **solar panel and battery pack**, making it suitable for outdoor or remote installations where power lines are unavailable.
- **Voice and LED Notification Module:** Add a small **speaker** or **buzzer** with pre-set voice alerts (e.g., "Vehicle detected", "Free parking period exceeded") and LED indicators for system status or errors.
- **AI-Based Anomaly Detection:** Integrate an additional lightweight model to detect abnormal events — such as missing license plates, vehicle tailgating, or unauthorized parking — using pattern recognition.

- **Dashboard Web Interface:** Create a simple **web-based dashboard** (Flask or Streamlit) accessible over LAN/Wi-Fi, showing live camera feed, current detections, and database insights in real time.
- **Multilingual Support:** Add a configuration module that allows operators to switch between **English**, **German**, or other languages in the interface, making it suitable for broader deployment.
- **NFC-Based Admin Login:** Include a small NFC card reader for secure access to configuration and maintenance menus, ensuring only authorized staff can modify system parameters.

These proposed enhancements can transform the system from a fixed gate-level application into a flexible, intelligent, and scalable AI platform — suitable for airports, campuses, or smart-city environments.

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